

Analytical Framework Supplement

1. Methodology

a. The Challenge (Bronze layer)

I identified some opportunities of cleansing and improvement in the dataset:

- a. I found **1437 duplicate transactions**.
- b. **TransDate** column had **hours**, which is great for a higher granularity analysis, but since the date format of Winlostdate didn't, **I decided to stick with just days**.
- c. **All the money values had to be converted to SGD currency** for a homogeneous analysis.
- d. **The odds system found in the dataset was the Indonesian one**. I decided to convert it to decimal odds to avoid having negatives values and be able to create groups/segments that help me outline the clients.
- e. BetGameTime column had empty values when LiveIndicator was false, which makes sense, but **to avoid using two columns for one measurement**, I had to fill those blanks with a value ("No Live").
- f. **The data type of every column was changed** to match the ones in the definition tab of the assignment.

b. The Solution (Silver to Gold)

Once the data was cleansed and silver layer ready, **it was time to add some business value to get it to gold layer level** and help me better understand the situation of the case and generate better insights. Based on that, I decided to add the next columns:

- a. **CodMonth**: Even when I only had two months in the dataset, it is important to have easy column to distinct the month. Also, when doing average values in DAX, this column helps a lot.
- b. **BetTypeName**: For faster understanding and most important, for presentation purposes, I just added the string names of every type of bet.
- c. **BetTypeGroup**: Based on the previous column, this aims to group them to have a better overall distinction.
- d. **BetTypeClass**: Same as before, but this was for distinct if the type of bet was for the total match or for a team specifically.
- e. **Actual_Stake_SGD/Winlost_SGD/MaxBet_SGD**: The money values converted to SGD currency.
- f. **Decimal_Odds**: The Indonesian odds format converted to decimal odds.
- g. **Odds_Zone**: A segmentation based on how high or low are the odds our clients tend to bet on.
- h. **Client_Segment**: A segmentation for our clients, based on much they bet (actual stake).
- i. **LeagueName**: Name of the league based on the code and the definition tab.
- j. **WebName**: Same as the previous case.

- k. **Flags:** Extra values for filtering the data based on bet limit, valid bet and the casuistry of losing a bet and not losing money.
- l. **TimeBetResultDays:** Difference in days between when the bet was made and the result.
- m. **WaitingTime:** Base on the previous point, I made an aggregation base on how many weeks it took for the result or if it was on the same day.
- n. **BetGameTimeGroup:** This is an aggregation for when exactly our clients bet. Did they bet before the match, during, and if during, first half or second half.
- o. **Country and Continent:** Based on the currency.
- p. **Flags for churn:** To be able to understand what happened with football in march.

c. *How?*

To ensure my analysis was robust and clean, **I built a pipeline to get the original dataset to a gold layer level with Python in Google Colab.** Here you can see the step by step:

Notebook: <https://colab.research.google.com/drive/1WxGqF3Y8JxqOs8Gx2TKkNZCXJ-16LUgV?usp=sharing>

The process involved cleaning, deduplication and aggregating **to prepare the data for the Power BI model.**

d. *Objective (Why?)*

With an improved dataset, it was time to **build a dashboard that would allow me to get a macro to micro view of the data** and also, **identifying insights** and formulate strategic recommendations based on **opportunities for improvement.**

But in order to start from the top, **I needed to define the KPIs that would help me go flawlessly from the top to bottom and tell a good story.** Based on the business case, we are working with Sports Betting Websites business, so I thought these will help me the most identify if we are doing good or not, financially:

- I. **Gross Gaming Revenue (GGR)** – Since I have no data of costs, I will be using this instead of the net. It is calculated by **Total Stakes minus Total Payouts**, which I find in the provided dataset, once they are converted to SGD. **It tells me if I am actually having profit** before subtracting costs. This is the first step of the funnel.
- II. **Hold (%)** – This acts as a **conversion rate** if we see it from the sales or marketing perspective. It is calculated by **GGR divided by Total Stakes**. This basically tells me **how much of the money getting in I am able to keep as profit**. A healthy range for this kind of business is 7-12%.
- III. **Average Revenue Per User (ARPU)** – I see this as a productivity measurement. How are my clients performing in my websites? It is calculated by **dividing GGR by number of unique clients**. This tells me **how much profit I gain per user in average**. This kpi is very useful to have a preview if some website or segment of clients could be churning or needs an adjustment for the allowed players.

With this in mind, **I will be able to answer the questions of the assignment correctly** by building a Power BI dashboard, since it is easier to work there with this kind of amount of data.

2. Sales Summary

Since in march there was a total absence of football bets, I will be focusing only on April performance. I am also not considering invalid bets based on the definitions.

a. High level takeaways

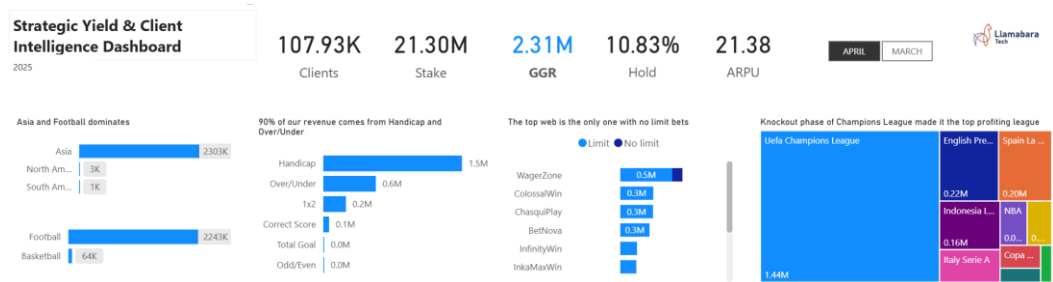
- a. **107.93K clients** bet on April. **99% was in Football and from Asia**, as expected.
- b. These clients wagered a total of **21.3M SGD**.
- c. We were able to **hold 2.31M (GGR)** of the total stakes. That gave us a **hold percentage of 10.83%**, which is in the healthy range for this kind of business (7-12%).
- d. Our overall **ARPU is at 21.38**. Giving the most of our bets are from Asia, **a healthy range for this kpi falls between 15-30**, which is great. Giving that these are high volume, low margin regions, the average ticket size is small.

b. Drill-down takeaways

- a. **90% of our revenue** comes from **Handicap and Over/Under** bets.
- b. The most **profitable** websites are **WagerZone, ColossalWin, ChasquiPlay and BetNova**, which all together brought **60% of the total profit**.
- c. On the other side, and not specifically looking at the bottom, but at their hold percentage, **the least profitable are PachaWin (6.75%), FastStake (5.99%) and SureStrike (6.21%)**. They fall behind the healthy range, but we will be looking at those in the recommendations section.
- d. Since the **Champions League** is almost at its peak in April with its **knockout phase**, it makes sense that it **was the top league in number of bets and GGR**.
- e. Based on only GGR, I would tell you that the less profitable league is La Liga 1 from Peru, but it has a good hold percentage (12.41%). Even with a higher GGR value, the **worst league is the Primera Division from Mexico**, with the lowest KPIs performance.

c. Visuals

I am adding here the sales part of the dashboard I made for all the assignment. It is being attached to the mail response for you to play with it.



3. Core Insights

Before stating my insights, I would like to highlight the **segmentation of clients** I made based on how much they bet on our websites, because this will help understand my findings better:

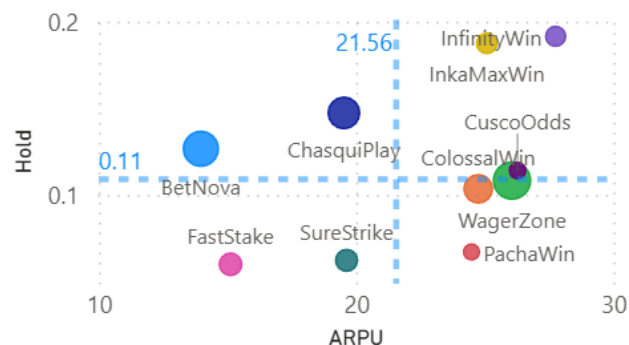
- **Micro/Test:** 0 - 10 SGD
 - **Casual:** 11 - 100 SGD
 - **Serious:** 101 - 500 SGD
 - **High-Stakes:** 501 - 2000 SGD
 - **Whale:** 2001 - 6000 SGD
 - **Outlier:** +6000 SGD
- In march, **we had only 305 clients for Football**, comparing to almost 107K in April. Also, there were literally no bets made during the matches (live). Since I have no past data and **April's data shows that most of football bets are made live (57%)**, I am assuming there was some **technical problem with live betting for football** and clients didn't want to bet that way, because a lot of leagues kept going strong during march. If this is the case, **we didn't only lost revenue that month, but clients in the future.**
 - In addition to the low number of wagers in March, **Football had all its KPIs in negative, unlike April.** This is explained by the type of clients it had. Even when the winning ratio in that month wasn't high, **all the lost bets (404) came from Casual clients (7.21%)**, which means low stakes low revenue. On the other side, **winning bets (155)**, even being lower than the lost one, they had a higher average ticket, because **they came from Whale (41%), Serious (14.72%) and Casual (20.23%)**. We could have ended up losing only 381.90 SGD, but that's without the costs.
 - Basketball, while even increasing its client's number and GGR, **reduced its hold kpi by almost 21 percentage points** from March to April. This was because of the clients as well. **The winning ratio went from 46.47% in March to 73.84% in April.** Also, Serious and High-Stakes stakes' share grew from 31.06% to 39.44% and from 17.64% to 24.23% respectively. **More winning bets from sharp clients hurt directly our Hold KPI.**
 - Even when **Basketball** is just the 3% of the total clients who bet in April. It managed to keep 86.65% of the clients from march, **achieving a churn of 13.35%.** Most of the websites in March had good ARPU and Hold values, which are good indicators of retention, so I assume these are more than 6 months old clients.
 - Most of our bets are made during the first half (34.35%), **but the most profitable are the ones made before the matches.** A hold of 14% vs 8.81% of live ones. Also, the winning ratio is lower with a 48.97% vs 62.41%. **It's the perfect combination for a higher GGR.**

- f. The winning ratio is our strongest enemy, because it affects our Hold directly and even more if sharp clients lead the way. **In April, we saw variation from 44.66% to 59.05%.**
- g. **Several websites have a good ARPU**, which is a great indicator of good client's retention, but **some are affected by the winning ratio and can't hold its real potential.** For example, ColossalWin and PachaWin.
- h. Based on the data, **what will make us more profit (GGR) is the combination of healthy ARPU and HOLD.**

4. Strategic Recommendations

Based on the analyzed data, here are my best recommendations:

- a. Giving that our winning ratio is bigger in April, **we should look to reduce it by managing better our odds.** As the data shows, higher the odds, lower the ratio. But we should find a balance, no client wants to lose most of the time.
- b. We should have an **aggressive targeted campaign to get more clients to bet before the matches.** It is our most profitable time (14% Hold). Maybe start with the new clients for easier education.
- c. A healthy monthly churn rate for old clients should be no more than 20% in this type of business. Basketball had 13.35%, led by good hold and arpu values. **These should be our main priority to increase GGR and reduce the churn.**
- d. Based on the previous point, **I strongly suggest to work first with the websites FastStake and SureStrike.** If you take a look at the image, they are in the worst quadrant (lowest hold and lowest arpu). **This means high probability of churn in the next month and a poor revenue.** Our goal is to eventually move all our websites to second quadrant, where InfinityWin and IkaMaxWin are.



5. Predictive Analysis

Since I have no past data to get a better feeling of a trend, and based on real world behavior of the Premier League, these are my assumptions:

1. May is the last month of the Premier League, so there is a probability that the champion is defined that month. **More viewership, more bets, but not necessarily more money bet.**

2. At the same time, there is also a probability that the champion is defined at the end of April. This means **less viewership on May, but not necessarily less money bet, since the money can be distributed to other “safer” bets.**
3. There is always a date at the end of may where matches are played simultaneously. This day is always a great peak **which can increase or reduce our overall hold percentage. Generally, it translates into +1% or +2%.**
4. So far, our stake and hold are not that safe to bet on for a prediction. So, to have a good support, I checked a “Market Impact” report from **Gambling Commission** (<https://www.gamblingcommission.gov.uk/news/article/market-impact-data-on-gambling-behaviour-operator-data-to-sept-2025>), which tells us that **at the end of quarters of the year, active accounts or users tend to drop around 7-14%, but at the same time, the revenue can be higher with less users.**
5. Based on the source I will assume our **number of clients will drop by 14%** from April to May, but the **Stake Per User will maintain from April.**

Currency	April			May	
	Stake	Clients	Stake Per User	Clients	Stake
IDR	1,825,943	23,449	78	20,166	1,570,311
THB	1,824,365	13,563	135	11,664	1,568,954
VND	173,447	1,195	145	1,028	149,164
USD	42,321	232	182	200	36,396
HKD	9,841	16	615	14	8,463
CNY	9,251	43	215	37	7,956
MYR	8,470	23	368	20	7,285
MMK	7,004	85	82	73	6,023
LAK	4,424	26	170	22	3,805
IDO	3,235	72	45	62	2,782
VNO	1,354	4	338	3	1,164
No Currency	477	20	24	17	410
BRL	390	30	13	26	335
AUD	87	1	87	1	75
BDT	3	1	3	1	2
NGN	1	4	0	3	1
Total	3,910,611	38,764	101	33,337	3,363,126

As I was saying, the clients dropped by 14% from April, but the Stake Per User stayed the same. The Final Stake (SGD) was calculated by multiplying my new clients with Stake Per User value. **I am predicting May with less total stake**, but if our hold percentage increases, **we can have a better GGR than April.**